Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE
Course Code & Title: EE8351 Digital Logic Circuits
Name of the Faculty member: Ms. S. Sharmila Kumari, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Digital logic family
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO4
- **Activity Chosen: One minute paper**
- **Justification:**
  - The topic Digital Logic Family has 5 types - RTL, DTL, TTL, ECL and MOS families since each type has its own characteristics and operations. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of families which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic withoutwriting anything.

Total Strength is **31**

Reporter: **Myself**

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of digital logic family -3 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)

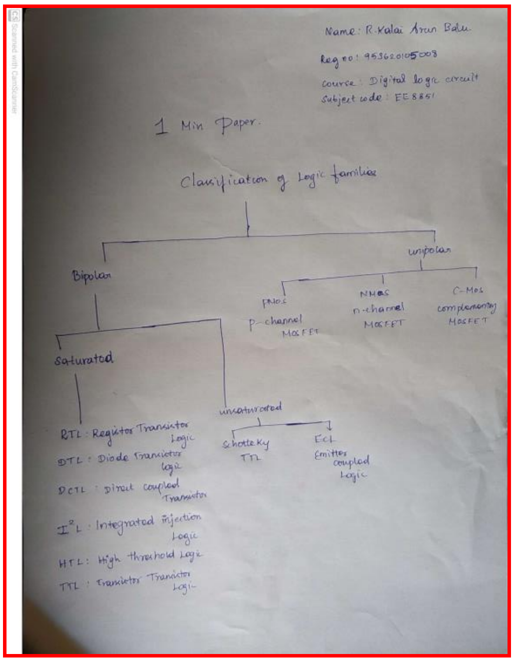
- **CO – PO / PSO mapping:**

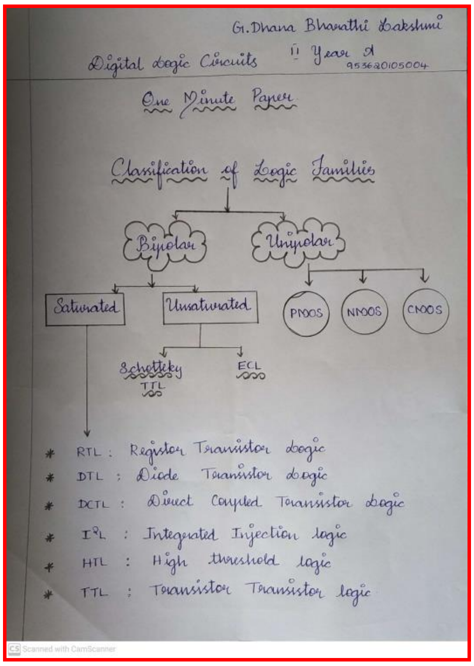
CO	PO1	PO2	PO3	PO10	PO12	PSO2	PSO3
C202.1	3	2	1	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO10**	**PO12**	**PSO2**	**PSO3**
	3	**2**	**1**	**1**	**1**	**1**	**1**
Justification for correlation	Students will have fundamental Electronic engineering concepts to solve digital circuit problems.	Students will identify the mathematical, engineering and other relevant knowledge that applies to a digital system problem.	Students will determine the design objectives and functional requirements of the digital logic family circuits.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the digital logic fundamentals in the advancement of VLSI technology	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with digital skills required for industrial automation systems.

- **Images / Screenshot of the practice:**





❖ *Reflective Critique:*

1. **Pre-implementation Reflection:**

   Benefits:

   - Students can able to attend the question even in the questions are in indirect form.
   - Students can able to explain the concepts in examination without any confusion.

   Challenges:

   - In the class mostly boys hesitate to answer the questions.
   - Time utilization for conducting activity.

2. **Post-implementation Reflection :**

   Benefits:

   - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

   Challenges:

   - Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that the students will be able to explain various types of logic family.
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 80% of students answered.

References:

❖ https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-one-minute-paper

Signature of Faculty Member **HOD**

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Innovative Practice Description

- **Unit / Topic:** Unit II / Multiplexers and De-multiplexers
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen: Mini - map**
- **Justification:**
  - Differentiate the multiplexers and demultiplexers
  - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the multiplexers and demultiplexers which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

- **Details of the Implementation:**

After teaching the concept, the students were made to pair with their neighbors
Reporter: Myself
At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about multiplexers and demultiplexers concept for 2 minute.
- ✓ Then I told them to Pair with their neighbours and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design mini-map and submit within 3 minutes.

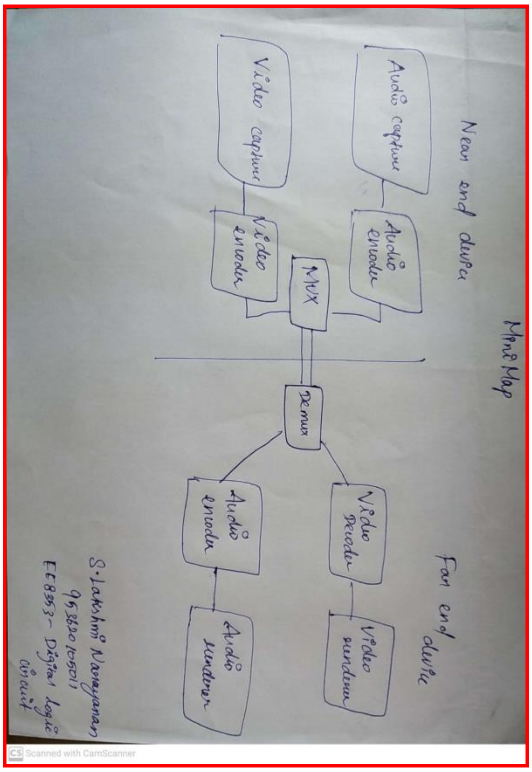
- **CO – PO / PSO mapping:**

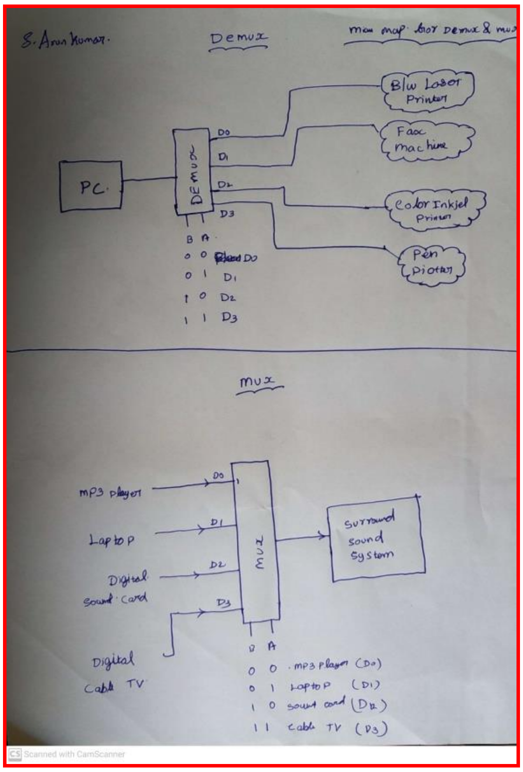
CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO2	PSO3
C202.2	**3**	**2**	**2**	**3**	**1**	**1**	**1**	**1**

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO10**	**PO12**	**PSO2**	**PSO3**
	3	**2**	**2**	**3**	**1**	**1**	**1**	**1**
Justification for correlation	Students will have fundamental Electronic engineering concepts to solve digital circuit problems.	Students will identify the mathematical, engineering and other relevant knowledge that applies to a digital system problem.	Students will determine the design objectives and functional requirements of the combinational logic circuits.	Students will use data in tabular and/or graphical forms methods including design of experiments of combinational logic circuits.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the digital logic fundamentals in the advancement of VLSI technology	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with digital skills required for industrial automation systems.

- **Images / Screenshot of the practice:**





- ❖ *Reflective Critique:*

  1. **Pre-implementation Reflection:**

     I preferred this Activity because they have different types of combinational circuits separately.

     Challenges anticipated:

     - ❖ In the class mostly boys hastate to answer to the questions.
     - ❖ Time utilization for conducting activity.

     Steps taken:

     - ❖ The boys are sitting in 2 columns – I planned to choose more pairs from boysto involve them in the activity.

  2. **Post implementation Reflection:**

  - **Benefits:**
    - ➢ Students understood the concept which was reflected from their answers forthe questions I have asked during discussion session.
  - **Challenges:**
    - ➢ Slow learners were not able to understand some topics during discussionhours.

- ❖ *Benefit of the practice:*

  - The assessment of effectiveness of the activity was felt when told most of the points.
  - While conducting the activity, I understood that the students will be able to explainthe concepts of multiplexers and demultiplexers.
  - The success of the activity was evaluated by asking the same question in InternalAssessment test I – Around 50% of students answered.

References:

1. https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps

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Innovative Practice Description

- **Unit / Topic:** Unit III / Shift Registers
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen: Think Pair Share**

- **Justification:**

Differentiate types of Shift Registers
After teaching the concept, I thought of conducting this activity for making the students to give the difference between the types of Shift Registers which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

- **Details of the Implementation:**

After teaching the concept, the students were made to pair with their neighbours (i.e) In a Class.

Total Strength is 31, Number of Pairs – 10
Photographer: one student – Mr Aswath Arjun (interested in photography)
Reporter: Myself
At the end the Class (Last 6 minutes)

- I asked the students to **think** about Shift Register and its types concept for 2 minute.
- Then I told them to **Pair** with their neighbours and discuss about the difference between the types for another 1 minute.
- Finally, I selected on Pair from each column randomly and ask them to **share** one difference between the concepts. (3 miuntes)
- When each pair told the points simultaneously, I wrote the points on the board.
- Finally, I summarized the points and I told one missed point to all the students.

- **CO – PO / PSO mapping:**

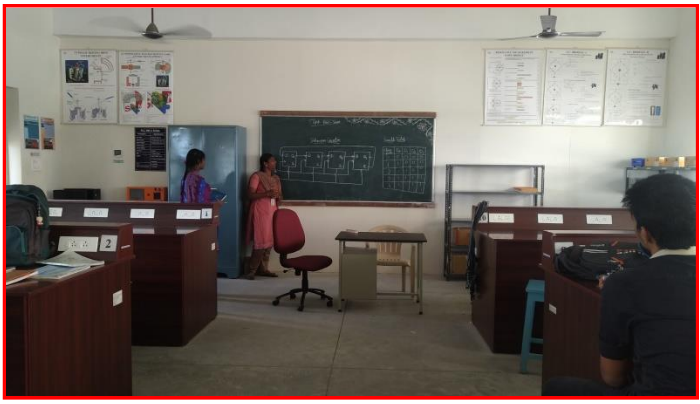
CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO2	PSO3
C202.3	**3**	**2**	**2**	**3**	**1**	**1**	**1**	**1**	**2**

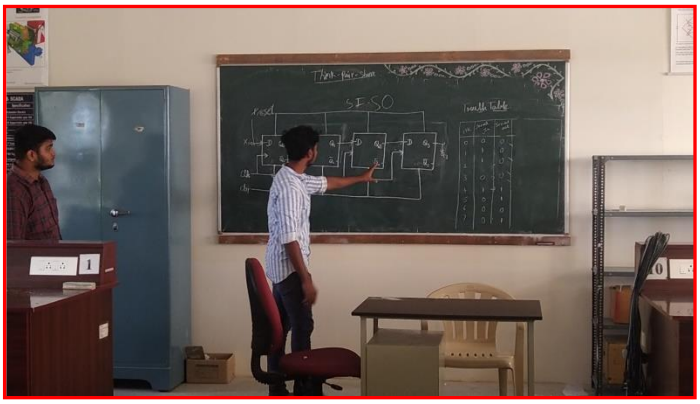
- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO9**	**PO10**	**PO12**	**PSO2**	**PSO3**
	3	**2**	**2**	**3**	**1**	**1**	**1**	**1**	**2**
Justification for correlation	Students will have fundamental Electronic engineering concepts to solve digital circuit problems.	Students will identify the design specification of synchronous the sequential circuits.	Students will determine the design objectives and functional requirements of the synchronous sequential circuits.	Students will use data in tabular and/or graphical forms methods including design of experiments of sequential logic circuits.	Students will be able to work with their team through Active Learning Methods.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the digital logic fundamentals in the advancement of VLSI technology	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with digital skills required for industrial automation systems.

- **Images / Screenshot of the practice:**

A ***view of implementation of the Activity*** while the students are sharing the points





❖ *Reflective Critique:*

1. **Pre-implementation Reflection:**

I preferred this Activity because they have different types of Shift Registers separately.

Challenges anticipated:

- In the class mostly boys hastate to answer to the questions.
- Time utilization for conducting activity.

Steps taken:

- The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. **Post implementation Reflection:**

Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

Challenges:

- Slow learners were not able to understand some topics during discussion hours.

❖ *Benefit of the practice***:**

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that one point was missed with respect to method of computation and finally I insisted the point they missed to tell.
- The success of the activity was evaluated by asking the same question in **Internal Assessment test II – Around 60% of students answered correctly.**

References:

1. http://www.readwritethink.org/professional-development/strategy-guides/using-think- pair-share-30626.html
2. https://in.video.search.yahoo.com – Think pair share

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Innovative Practice Description

- **Unit / Topic:** Unit IV / PROM, PLA and PAL
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen: Flipped Classroom**
- **Justification:**

Flipped classroom is a "pedagogical approach in which direct instruction moves from the group learning space to the individual learning space, and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creatively in the subject matter"

Time Allotted for the Activity: 30 Minutes

- **Details of the Implementation:**

Number of Students – 4

Photographer: one student – Mr. A.S.Kannan (interested in photography)

Reporter: Myself

At the Class (30 minutes)

- ✓ I shared the material related to the topic through canvas LMS.
- ✓ Then I told them to prepare the topic as a team with a help of PPT.

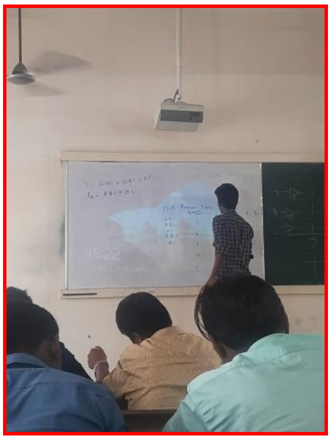
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO2	PSO3
C202.4	3	2	2	3	1	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO2	PSO3
	3	2	2	3	1	1	1	1	2
Justification for correlation	Students will have fundamental Electronic engineering concepts to solve digital circuit problems.	Students will identify the design specification of Asynchronous sequential circuits and Programmable Logic Devices.	Students will determine the design objectives and functional requirements of the Asynchronous sequential circuits and Programmable Logic Devices.	Students will use data in tabular and/or graphical forms methods including design of experiments of sequential logic circuits.	Students will be able to work with their team through Active Learning Methods.	Students will be able to communicate the technical concept through Active Learning Methods.	Students will use the digital logic fundamentals in the advancement of VLSI technology	Students will be able to design and test the electronic system in the engineering applications.	Students will be able to design and develop the hardware and software with digital skills required for industrial automation systems.

- **Images / Screenshot of the practice:**

- *Benefit of the practice:*
  - This practice improves the peer coaching / studying with in the class.
  - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

References:

1. https://www.panopto.com/blog/7-unique-flipped-classroom-models-right/

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